Flags, consoles, pagination, master data

The four delivery questions LORA's delivery.md asks — can you choose what deploys, is custom UI hard, is pagination solved, can master data feed dropdowns — asked of Bravo, plus what production says about the front ends.

5	**15+**	**2.41M**	**1+1+0**
WAYS TO PICK WHAT RUNS — AND NO REGISTRY FOR ANY OF THEM	BROWSER APPLICATIONS IN THE BRAVO ORBIT. LORA HAS TWO	CONSOLE ERRORS IN 7 DAYS — 5.8× LORA'S BACK OFFICE	ONE YAML LINE, ONE SQL UPDATE, ZERO JAVA — TO RENAME AN APPROVAL ROLE

Audience Platform, front-end engineers, product engineers shipping DF and NDF work **Sources** `bravo-bpm-service` v2.93.43 · Datadog us5 RUM and APM **Read** 2026-09-10

The four questions, plus the one nobody asked

Can you pick what deploys?	● YES — FIVE WAYS	**Which is the problem.** Bravo has no LaunchDarkly either, but it has Spring property flags read *inside BPMN gateway strings* (78 lookups), a six-column database selector, a per-application jsonb activity on/off matrix, Camunda definition versioning, and 31 Java factories. **No registry, no owner, no expiry.** Toggling a gateway flag needs a configuration change *and a restart*, because it is `environment.getProperty` evaluated at gateway time.
Is a custom front end on a page hard?	● NO — AND THAT IS THE TRADE	Every Bravo console is an ordinary React app calling domain REST verbs; the UI never sees a Camunda task id. **One page is easy.** But there are **at least fifteen** separate browser applications, and a change to a shared concept ships in as many of them as touch it. LORA has one back-office front end and a three-layer widget seam. **Bravo traded a hard seam for a wide surface.**
Is pagination solved?	● YES — AND LORA'S ACCOUNT IS CONFIRMED	From Bravo's own telemetry. Bravo built exactly the paging LORA describes: a controller literally named `SurveyorAssignmentFormTabV2Controller`, plus **nine** per-assignment sub-resource endpoints. Together they are the **busiest read path in the service.** LORA is right not to rebuild it — and Bravo is right that it needed it.
Can master data feed dropdowns?	● YES — THE CAPABILITY LORA DECLINED	LOV tables and three master-data services back the dropdowns directly, which is why renaming an approval role (`BLCS-4683`, NMH→GMB) took **one YAML line, one SQL `UPDATE` and zero Java**. The cost is the one LORA designed around: values are read *live*, so a mid-flight master-data edit can change what a form offers *after a decision has been taken on the old list.*
Are the front ends healthy?	● NO, AND NOBODY WAS LOOKING	Four instrumented consoles produce **2,413,198 errors in 7 days** — 5.8× LORA's back office. **748,686 of them are 404s** on the Surveyor Platform, one endpoint reaching **69% of all sessions. No monitor can see it because every health check filters on `5*`.**

> **Every delivery mechanism Bravo needs exists, and most of them work. What is missing is a boundary.**
>
> No flag registry, no console inventory, no front-end error budget — so the delivery surface grows and nothing tells anyone it has.

Feature flags: five mechanisms, no registry, and a restart

LORA's complaint was "no feature flag; cannot pick what deploys", graded *partially true*. **Bravo's answer is the opposite failure.** It has five ways to pick what runs, and they are not coordinated.

MECHANISM	GRANULARITY	TO CHANGE IT YOU…	IN-FLIGHT LOANS?
Spring property flags inside BPMN gateway conditions — `environment.getProperty('setting.feature.config.featDF2W') == 'true'`**; 68 lookups in the legacy monoliths, 10 in the unified spine**	environment-wide	change configuration and **restart**, or refresh config	yes, at the next gateway evaluation
Java service flags — `featRefactorWorkflow`, `featFinalApproverRoleOnReject`, `featSingleApproverRejectedRole`	environment-wide	same	yes
Six-column selector tables — `WorkflowProductConfig(productId, type, customerType, userType, businessType, riskType)`	per product / customer / risk combination	a Flyway **data migration**	at resolution time
Per-application jsonb activity matrix — `{"CI": {"PilotBranchCheckActivity": [true,false]}}`, **read by** `BaseActivity.execute()` **before every unified activity**	**per application, per activity**	resolved at start, re-resolved mid-flight for the "SUO" phase	fixed per application at start
Camunda definition versioning	per process definition	deploy	parent stays on its version; **children resolve to latest at call time**

WHAT IS GENUINELY GOOD HERE

The jsonb matrix is a **real per-application guard layer sitting on top of the flowchart** — the closest thing in either platform to LORA's `$.experiments.*`, and it arrived independently. It can only *skip* a step the diagram already contains, never introduce or reorder one, and it is static per application rather than computed from data readiness — **but it is the right idea.**

A FLAG INSIDE A BPMN GATEWAY STRING IS NOT A RELEASE TOGGLE

Flipping it is a **configuration deploy, not a switch**, and it applies to the whole environment at once. That is a *slower* release control than a code change gated behind a property — because the property is now also in XML. And there is **no registry and no expiry**: 78 gateway lookups plus an unbounded set of Java flags, with no list of which are live. `featRefactorWorkflow` gates the entire unified-versus-legacy behaviour and has been in place since 2024.

BRAVO'S REAL DEPLOY RISK IS NOT FLAGS — IT IS CHILD-PROCESS BINDING

No `callActivity` sets `camunda:calledElementBinding`, so redeploying a sub-process changes behaviour for *running* loans at their next call — including NDF2W loans entering unified underwriting. **That is 73% of production volume with no migration plan.** It is the single most important "what deploys to whom" question on this platform.

Custom front end: easy per page, hard per system

LORA's complaint was that a custom widget is **three layers** — templ emits a custom tag, SolidJS registers the `customElement`, HTMX carries the value back — and that field-by-field templ cannot see sibling fields. Production bore it out: a script injected twice (8,067×/week) and hydration-order null dereferences (~34,900×/week).

BRAVO HAS NONE OF THAT SEAM

Each console is a plain React application talking to REST endpoints that speak **domain verbs** — `PATCH /v1/underwritings/{id}/approval/bm-decision`, `PUT …/approver-decision`. The UI never sees a Camunda task id or a form definition. **Adding a field, a control or a whole page is ordinary front-end work.**

THE COST IS ON THE OTHER AXIS: HOW MANY FRONT ENDS THERE ARE

Every shared concept — an approval banner, an asset-category field, a status label — is implemented and maintained **once per console that shows it**. The `BLCS` ticket stream shows the consequence directly: `BLCS-4397` *"Shared summary section config for both surfaces"* and `BLCS-4398` *"Verify section matrix across 5 flows on both surfaces"* are **one story spending its effort on keeping two consoles consistent.**

> **Per page, Bravo is clearly easier. Per system, Bravo pays a coordination tax LORA does not have —**
>
> and it pays it in exactly the place where its production error volume is concentrated.

CONSOLE, BRAVO/LOS ORBIT	RUM
Surveyor Platform (Prod + Test)	ALL
Operation Platform (Prod + Test)	ALL
Underwriting Platform (Prod + Test)	ALL
Backoffice Web App	ALL
Supplier Web App	ALL
Customer Platform 4W · 2W · Sharia · DF · unified	ALL
Agency Cockpit	NONE
`prod-bravo-e-line-ui`, `prod-notary-ui`, `prod-bravo-lms-fe`, `prod-bravo-lms-fund-fe`, `Prod LMS Core FE`, `prod-bfi-dashboard-digital-fe`, `prod-bravo-edoc-download`, `PROD-DMS-FE`, `EGC Platform`, agent and RO tooling	mixed

At least **fifteen** browser applications, where LORA has **two** — back office and customer.

Pagination: Bravo built it, it is called `FormTab`, and it is the busiest read path

CONTROLLER	SPANS / 2 D	WHAT IT IS
PartnershipConfigurationController.getByApplicationId	**73,056**	per-application configuration — and the 404 storm
SurveyorAssignmentFormTabV2Controller.findByAssignmentId	**33,673**	**the paged form itself, v2**
ApplicationDocumentTrackingController.get	32,820	documents pane
SurveyorAssignmentManualKycController.scoring	30,591	KYC pane
SurveyorAssignmentController.exposeSurveyData	8,535	survey payload
SurveyorAssignmentCompanyDataController.findByAssignmentId	7,526	company pane
SurveyorAssignmentCompanyLegalityDocumentController	7,476	legality pane
SurveyorAssignmentCapitalDetailController.findByAssignmentId	6,413	capital pane
LongSurveyController.findAllByApplicationId / findByAssignmentId	5,077 / 4,903	long-survey panes

Plus three more per-assignment sub-resources visible in the 404 table: `surveyor-assignment-other-business`, `surveyor-assignment-asset-validation-two-wheeler`, `surveyor-assignment-request`.

LORA's delivery document grades this complaint **misstated**, on this reasoning: *"In Bravo the surveyor saw all fields from all tasks on one giant form. They needed pagination by logical group. In LORA the surveyor opens one task and sees only that task's fields."* **That account is correct — and it can now be verified from Bravo's own production telemetry rather than from memory.**

A CONTROLLER NAMED `FORMTABV2` IS A PAGINATION MECHANISM THAT HAS ALREADY BEEN THROUGH ONE MAJOR VERSION

Nine or more sub-resource endpoints per assignment is "pagination by logical group" implemented as **one endpoint per group**. It works, it is heavily used, and it is the dominant read path in `ms-bpm`.

BRAVO'S DESIGN IS NOT A WORKAROUND

It is a consequence of the object it presents. A Bravo surveyor opens an *assignment*, which spans the whole application. **Paging it is the only sane thing to do. Calling this a defect misreads it.** And LORA's recommendation stands, now better evidenced: nobody should rebuild whole-application paging in LORA, because LORA's surveyor genuinely opens one task.

BUT IT HAS A REAL COST LORA DOES NOT PAY

Opening one assignment issues **roughly ten HTTP requests**, each a controller, each a set of JPA `repository.operation` spans against `prod-postgres-bpm-d2bpm` — which is 34.7% of Bravo's production Cloud SQL bill. And **four of those ten endpoints are the top four sources of the 404 storm**, so *the paging surface is also the failure surface.*

Master data: Bravo has the feature LORA declined to build

	BRAVO	LORA
Where dropdown options come from	LOV tables in the same database (`underwriting_job_level_lov`, `…_detail`, `surveyor_coverage`, `surveyor_level`) plus live calls to master-data services	process-global cache loaded at worker start from BFI DMS; JSON Schema `enum` / `lov`; widget APIs
Master-data traffic, 7 days	`prod-ms-product` 31,463 · `prod-ms-master` 27,954 · `prod-ms-branch` 22,784 · `prod-ms-employee` 64,725	~620k calls/week to `prod-ms-master` alone, via the gateway
Adding a new lookup	**insert rows; the form reads them**	LGS inner proxy + schema `lov` (+ optional cache key) — a multi-repo change
Renaming a role across the approval ladder	**one YAML line, one SQL `UPDATE`, zero Java** (`BLCS-4683`, NMH→GMB)	a schema change and a redeploy
Consistency for the life of the loan	**not guaranteed** — options are read live	**guaranteed** by copying values onto the document at write

THE `BLCS-4683` CASE IS THE STRONGEST SINGLE ARGUMENT FOR BRAVO'S APPROACH IN THIS PACK

Renaming an approval role across a *staged ladder* — the kind of change that ordinarily ripples through code — was a **configuration edit**, because the ladder is `underwriting_job_level_lov_detail` rows ordered by `global_level` (CAFH 200 → AMB 300 → CCU 400 → GMB 600) snapshotted onto approver rows. **That is data-driven delivery working exactly as intended.**

AND THE COST IS EXACTLY THE ONE LORA DESIGNED AROUND

Because options are read *live*, a master-data edit changes what a form offers immediately — including for applications already in flight and **already decided on the old list.** Bravo mitigates this partially: the approver ladder *is* snapshotted onto approver rows at assignment. But that is **one path, done deliberately, not a platform property.**

A LIVE ILLUSTRATION OF THE RISK, IN THE TICKET STREAM

`BLCS-4683` renamed NMH→GMB in September 2026. `BLCS-4811`, **three days before this document**, is *"[BE] Prepare Query Change GMB to NMH"* — **the rename is being reversed.** A configuration change that is cheap to make is also cheap to make twice, and each edit lands on whatever applications are in flight at that moment. **Nothing in Bravo records which applications were decided under which version of the ladder**, because the master-data tables have no temporal dimension and Camunda's history purges at 90 days.

Neither design is wrong. Bravo optimised for change velocity on slowly-changing reference data and got it; LORA optimised for per-loan immutability and pays a multi-repo change for every new lookup. The right answer differs per lookup — and neither platform lets you choose per lookup.

2.4 million console errors a week, and nobody had looked

Sections 1–4 are code and configuration reading. **They did not have to be.** All four production Bravo LOS consoles are RUM-instrumented at `rum_event_processing_state: ALL` — and, as with LORA's back office, nobody had opened the view.

CONSOLE	SESSIONS	VIEWS	ACTIONS	ERRORS	ERR/VIEW
Surveyor Platform Prod	79,090	953,115	3,739,210	1,962,909	2.06
Operation Platform Prod	17,135	96,707	2,516,533	287,646	2.97
Underwriting Platform Prod	15,628	205,666	1,964,283	124,422	0.60
Customer Platform DF	2,594	31,412	87,473	38,221	1.22
Total, 7 days	**114,447**	**1,286,900**	**8,307,499**	**2,413,198**	**1.88**

For scale: LORA's back office produces 416,201 errors a week at ≈2.8/view over 24,172 sessions. **Bravo's LOS consoles produce 5.8× the error volume at a comparable per-view rate. Both teams had the instrumentation and neither was reading it.**

TOP ERRORS, SURVEYOR PLATFORM	/ 7 D
`Request failed with status code 404`	748,686
`csp_violation` — `cdn-cgi/rum` blocked by `connect-src`	35,415
`intervention: Ignored attempt to cancel a touchmove event` — tablet devices	20,928
`Request failed with status code 400`	12,796
same beacon, explicit-port origin	10,573
`Error getting location: "[GeolocationPositionError]"`	10,365
`Unable to get current position` — same defect, second message	10,325
`Request failed with status code 401` — the browser face of 8,533 IAM 401s	7,402
`Network Error`	6,828
`Cannot read properties of null (reading 'filter')` — real null dereference	6,445

The console behind `Surveyor Platform - Release reject` — **28.4% of Bravo's entire support-ticket load.**

Caveats. RUM sessions are sampled and the counts include benign browser noise — the two CSP rows (45,988 combined) are a third-party beacon and are cosmetic, and `utc_offset is deprecated` (5,950) is a library warning. Treat the 404 rows, the geolocation rows and the null dereference as the actionable ones. Window is 7 days to 2026-09-10.

The paging surface is the failure surface

A · THE 404S ARE THE PAGING SURFACE FROM §3

ENDPOINT	404S / 7 D	SESSIONS
/bpm/v1/partnership-configuration/feature-configuration	266,767	54,350 of 79,090 — 69%
/bpm/v2/surveyor-assignment-form-tab/assignment/{id}	157,623	37,359
/bpm/v1/application-document-tracking	154,097	45,916
/bpm/v1/surveyor-assignment-manual-kyc/scoring/assignment/{id}	143,180	31,565
/surveyor/assignment-detail/{id}	21,125	14,754
/surveyor/assignment	14,565	11,287

Three of the top four are the assignment sub-resource endpoints — the `FormTab` paging model. The first is the busiest handler in the entire service, and **two-thirds of all surveyor sessions receive a 404 from it.** Whether that is benign (an optional configuration probe where 404 means "no override") or a defect is a question for the controller, not for telemetry — but at this volume, on this console, it has to be asked. *It is section 3's mechanism and section 5's largest error, and nobody had connected them because nobody had opened either view.*

B

· GEOLOCATION FAILS ~20,700 TIMES A WEEK ON A PLATFORM SPENDING RP64.0M/MONTH ON MAPS

`GeolocationPositionError` and `Unable to get current position` are **the same defect reported twice.** Surveyors are field staff whose submissions depend on position capture. Separately, bravo-cost.md §5 finds Geocoding, Places and Maps billing **Rp63,979,414 in August**. This document does not claim the two are causally linked — a browser geolocation failure and a server-side Maps call are different layers — **but they are the same feature, in the same console, and neither has been examined.**

C · NO ALERT CAN SEE ANY OF THIS

```
(hits - hits{http.status_code:5*}) / hits < 99
```

A 404 is a *success* by that definition, and **none of the 38 monitors touches RUM at all.** This is the delivery-side face of the same finding: **Bravo can ship a console change that 404s two-thirds of sessions and every signal it owns stays green.**

HOW TO REPRODUCE

```
# Datadog RUM, Surveyor Platform Prod
@application.id:2f3ca103-2053-4601-8e2a-5de3aee713bb
  @type:error
  → group by @error.message

# the 404 endpoints
@application.id:2f3ca103-… @type:resource
  @resource.status_code:404
  → group by @resource.url_path_group,
    count(*) and cardinality(@session.id)
```

Nine actions

1 Read the `feature-configuration` 404 · S, DO THIS FIRST

266,767 a week, 69% of surveyor sessions, on the busiest handler in the service. Decide whether 404 is the intended “no override” response — and if it is,

stop the client raising it as an error

, because it is currently drowning that console’s error stream. If it is not, this is a live production defect that has been invisible for as long as anyone has data for.

2 Fix the geolocation failures, and check them against the Maps bill · S–M

~20,700 failures a week on a field-staff console. Establish the cause (permissions, HTTPS context, device, timeout) and whether the retries are billed against the Rp64.0M/month Maps line.

3 Set a front-end error budget per console · S

Today the Surveyor Platform runs at 2.06 errors per view and Operation at 2.97, and

no number would move if either doubled

. One SLO per console, reviewed weekly, is the cheapest thing on this list.

4 Filter the CSP beacon rows · S

45,988 `cdn-cgi/rum` violations a week are cosmetic and are the second- and fifth-largest rows on the Surveyor Platform. Either allow the origin in `connect-src` or stop loading the beacon;

right now they hide real errors.

5 Build a feature-flag registry · S–M

78 gateway lookups plus the Java flags plus the selector tables, with no list of what is live, what it gates, who owns it, or when it can be removed.

Start by listing them; the removal candidates will be obvious.

6 Pin `calledElementBinding`, or accept in-flight child migration deliberately · S–M

This is the real “choose what deploys” question on Bravo and it is

currently answered by default.

Cross-referenced in bravo-testing.md recommendation 1.

7 Inventory the consoles and name a shared component owner · M

At least fifteen browser applications, with stories already spending their effort on “both surfaces”. Either a shared component library with an owner, or an explicit decision that each console diverges.

8 Give slowly-changing reference data a version, or a snapshot · M

The NMH→GMB rename and its in-progress reversal show a configuration lever cheap enough to pull twice. Applications decided under the old ladder are not distinguishable from those decided under the new one after 90 days. Either version the LOV tables or snapshot the values onto the application, as the approver rows already do.

9 Instrument Agency Cockpit, or retire it · S

It is the one LOS-adjacent console with `rum_event_processing_state: NONE`.

30, 60, 90 days

Nine recommendations, phased against **~25 person-days a month** of Squad S&U time and a slice of Platform/SRE.

SEQUENCING RULE

Fix what users hit before rebuilding what engineers dislike. The 404 storm reaches 69% of surveyor sessions on the console that generates 28.4% of Bravo's support tickets; the console inventory and the shared-component question are structurally important and *hurt nobody today*. So the live defects go first and the architecture of the delivery surface goes last.

SHARED WITH OTHER DOCUMENTS

Recommendation 1 is also **bravo-observability.md rec 1** and **bravo-cost.md rec 5**; recommendation 2 pairs with **bravo-cost.md rec 3**; recommendation 3 pairs with **bravo-observability.md rec 10**; recommendation 6 *is* **bravo-testing.md rec 1**. Phased identically everywhere — do them once.

Days 0–30

THE LIVE DEFECTS AND THE CHEAP NOISE

1 **Read the `feature-configuration` 404** — is 404 the intended "no override" response, or a defect? *S&U · 2 d.* A written answer for the endpoint hitting 54,350 of 79,090 sessions.

1 **Act on the answer** — resolve the lookup, or stop the client raising it as an error. *S&U · 3 d.* The Surveyor Platform's error stream drops by roughly a third, and whatever the 404 was hiding becomes visible.

4 **Filter or allowlist the CSP beacon** — 45,988 violations a week. *Platform · 1 d.* Real errors stop being buried.

9 **Instrument Agency Cockpit, or retire it.** *Platform · 1 d.*

6 **`calledElementBinding` investigation — read-only** *(shared). S&U · 5 d.* A written exposure note. **No code change.**

Days 31–60

BUDGETS, THE FIELD DEFECT, THE FLAG INVENTORY

2 **Fix the geolocation failures** — ~20,700 a week on a field-staff console; establish cause (permissions, HTTPS context, device, timeout). *S&U · 5 d.* Cause documented and cross-checked against the Rp64.0M/month Maps line.

3 **Set a front-end error budget per console** — one SLO each, reviewed weekly. *Platform/SRE + EM · 2 d.* A console regression is caught by a number instead of by 315 `Release reject` tickets a month.

5 **Build the feature-flag registry** — what is live, what it gates, who owns it, when it expires. *S&U · 4 d.* Somebody can answer "what is `featRefactorWorkflow` gating and can we delete it" without reading 53 BPMN files.

6 **`calledElementBinding` decision** *(shared). S&U + EM · 2 d.* A recorded decision with a migration plan.

Days 61–90

THE DELIVERY SURFACE ITSELF

6 **Implement the binding decision** *(shared; effort carried in bravo-testing.md)*. Redeploying a unified sub-process stops silently changing the path of in-flight NDF2W loans.

7 **Inventory the consoles and name a shared-component owner** — at least fifteen browser applications, with stories already spending their effort on "both surfaces". *EM + FE leads · 3 d.* Either a shared component library with an owner, or an explicit, recorded decision that consoles diverge.

Deferred

WITH TRIGGERS

7 **Shared component library**, beyond the inventory. An organisational decision with a budget, not an engineering task. *Trigger: after the phase-3 inventory.*

8 **Version or snapshot slowly-changing reference data.** 10 days, and the design depends on the Camunda history decision in **bravo-cost.md rec 4**, because both concern what is knowable after day 91. *Trigger: after that decision lands.*

What changes when this is done

RECOMMENDATION	EXAMPLE WHEN DONE
`feature-configuration` 404 resolved	The Surveyor Platform’s error stream drops by roughly a third, and whatever the 404 was hiding becomes visible.
Geolocation fixed	Field surveyors stop failing position capture ~20,700 times a week, and the Maps line gets an explanation.
Front-end error budgets	A console regression is caught by a number instead of by 315 `Release reject` tickets a month.
Flag registry	Somebody can answer “what is `featRefactorWorkflow` gating and can we delete it” without reading 53 BPMN files.
`calledElementBinding` decided	Redeploying a unified sub-process stops silently changing the path of in-flight NDF2W loans.
Console ownership	“Shared summary section config for both surfaces” is a component change, not a story.
Reference data versioned	The question “which ladder was this application approved under” has an answer at day 120.

> **Bravo can ship a console change that 404s two-thirds of sessions and every signal it owns stays green. One SLO per console fixes that.**